

A differentiable simulation is all you need: repressilator identification from sparse reporters

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1 Introduction

Inferring a gene circuit's parameters from sparse single-cell observations usually relies on **dedicated estimators**, optionally **with a simulation**. Using the repressilator circuit, we show that **backpropagation through a differentiable simulation itself matches or outperforms them**. What limits inference is the choice of reporters, which we assess with an identifiability analysis.

2 Circuit modeling

- Elowitz–Leibler repressilator [1], a model of an oscillating gene circuit, is extended with **three** transcriptional reporters on a p15A plasmid.
- Each reporter copies one circuit promoter and drives a **degradation-tagged fluorescent protein**.
- A repressor can itself be fused to a fluorescent protein.
- We model the **maturation rates** and the **titration effects** of reporters on repressors.
- One cell is simulated either as an **ODE** or as an **exact Gillespie process** and imaged every 5 minutes for 10 h total time.
- We estimate the circuit **parameters**, reporter strengths, leaks, and the **starting state**. Maturation, tag decay, and titration are fixed.

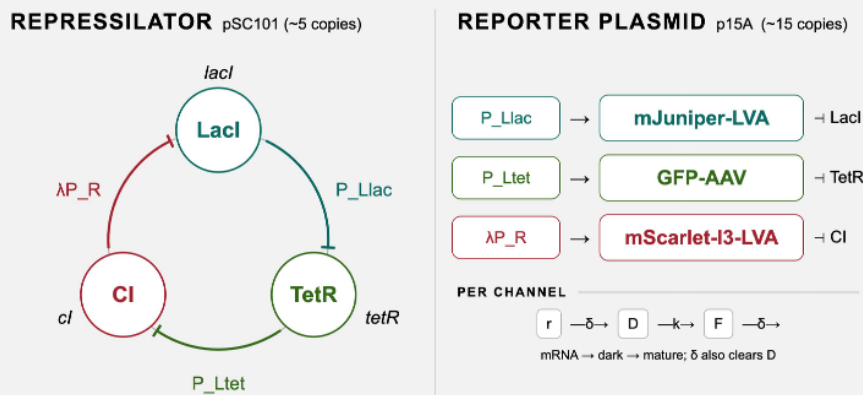


Figure 1: Schematic of the gene circuit used to benchmark various parameter inference methods.

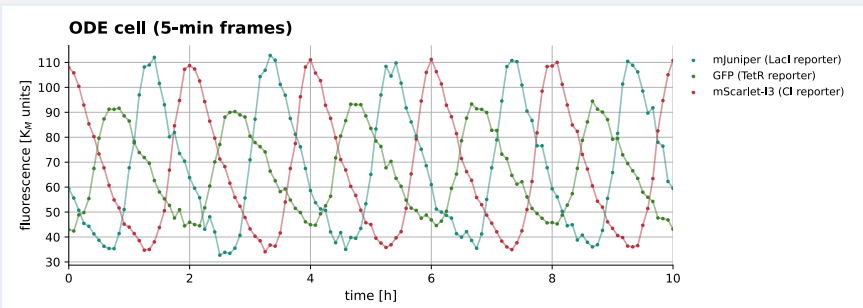


Figure 2: Reporter readouts over the course of 10 hours of simulated imaging using the ODE. Idealized case.

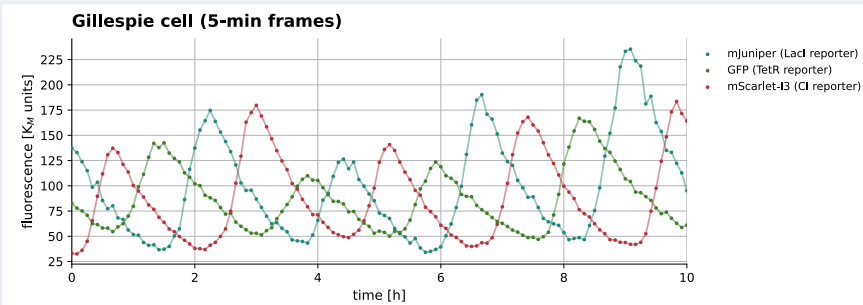


Figure 3: Reporter readouts over the course of 10 hours of simulated imaging using Gillespie's algorithm. Realistic case through additional phase drift over time and amplitude variations between consecutive cycles.

3 Differentiable simulation

- Governing ODEs are **smooth and differentiable** functions [2] of both the state and the parameters.
- Simulation implemented in PyTorch for its automatic differentiation, enabling **backpropagation-through-time** (BPTT) after simulation.

$$\frac{dm_{\text{TetR}}}{dt} = -m_{\text{TetR}} + \frac{\alpha}{1 + (p_{\text{LacI}} / K_{\text{LacI}})^r} + \alpha_0$$

$$\frac{dp_{\text{TetR}}}{dt} = -\beta(p_{\text{TetR}} - m_{\text{TetR}})$$

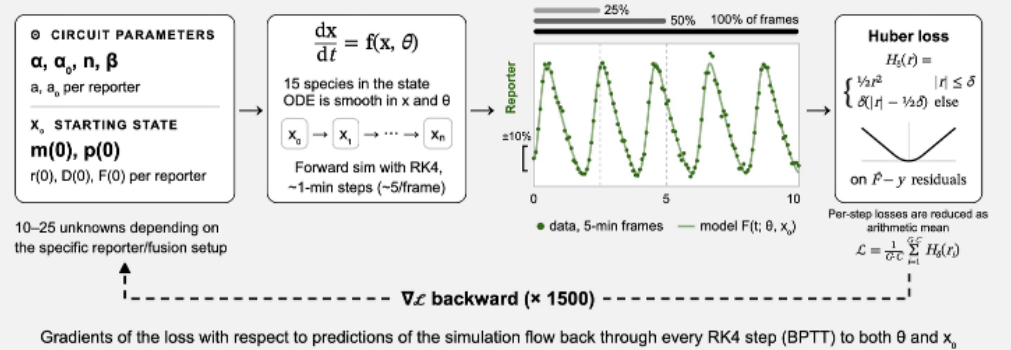
$$\frac{dr_{\text{GFP}}}{dt} = -r_{\text{GFP}} + \frac{a}{1 + (p_{\text{TetR}} / K_{\text{TetR}})^r} + a_0$$

$$\frac{dr_{\text{GFP}}}{dt} = \delta r_{\text{GFP}} - (k + \delta) D_{\text{GFP}}$$

$$\frac{dr_{\text{GFP}}}{dt} = k D_{\text{GFP}} - \delta F_{\text{GFP}}$$

4 Parameter Inference

- Trajectory shooting**: regress the initial state of the cell + parameters of the circuit. The rest is determined by the ODE.
- Randomized initialization** from biologically-justified intervals.
- Adam optimizer with Huber loss** for 1500 epochs, cosine learning rate schedule $0.05 \rightarrow 2.5 \cdot 10^{-4}$, increasing trajectory length curriculum.



5 Results

Our method was compared to recent, state-of-the-art estimators: ABC-SMC [3] (approximate Bayesian computation through sequential Monte Carlo), MAGI [4] (manifold-constrained Gaussian process Inference), an Inverse Physics-informed Neural Network [5] (neural network fitted to the ODE), and DGA [6] (differentiable Gillespie).

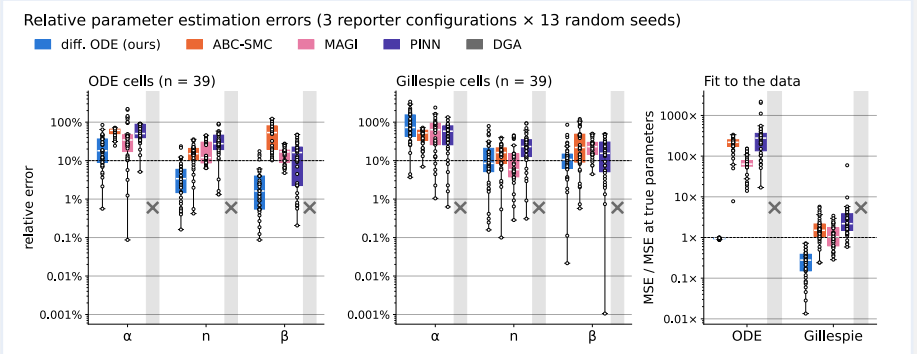


Figure 4: **Relative error** of the maximal transcription rate α , the Hill coefficient n and the protein-to-mRNA decay-rate ratio β , and fit to the data for each method, on **three reporter configurations** (GFP only, mJuniper, GFP and mScarlet-I3, TetR-FP fusion + GFP). Differentiable ODE (blue) often reaches the best possible fit.

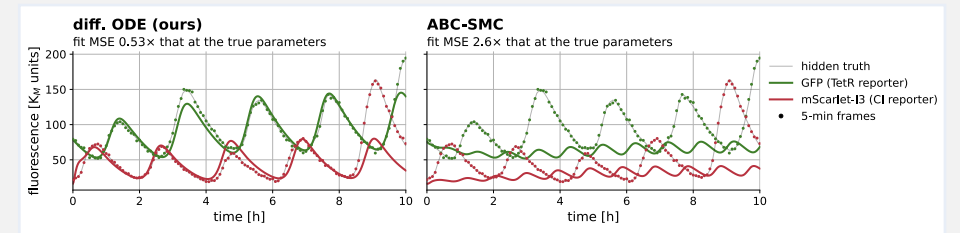


Figure 5: **One Gillespie cell with two reporters**, fitted by the differentiable ODE (left) and ABC-SMC (right). Dots: 5-min frames, lines: the ODE integrated at each. A fit MSE below $1\times$ means the fit matches this stochastic cell better than the **deterministic model at the true parameters** does.

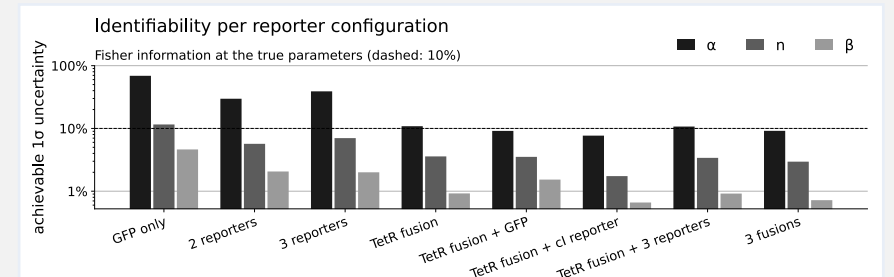


Figure 6: **Best achievable 1σ uncertainty** of each parameter for each reporter configuration, from the Fisher information of one cell's 5-min frames at the true parameters. Transcriptional reporters pin down the Hill coefficient n and the decay-rate ratio β , but α is only known relative to the unknown reporter strength. The leak a_0 (not shown) is unidentifiable in every configuration.

6 Conclusion

On the repressilator, **plain backpropagation through a differentiable ODE matches or outperforms dedicated estimators**. Accuracy is limited by the information revealed by transcriptional reporters, which constrain the circuit's timing but not its absolute transcription rate. Identifiability analysis can therefore guide reporter design before a physical experiment to **maximize chances of parameter recovery** and help select the most suitable inference method.

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